

# Alexandre Autran

Student in mathematics and machine learning

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## Profile

I am a mathematics and machine learning student at ENS Rennes and ENS Paris-Saclay, working in the mathematical foundations of machine learning and data science, with a particular focus on probabilistic methods, high-dimensional statistics, stochastic processes, and optimal transport.

## Education

- 2024–2027 Normalien – ENS Diploma in Mathematics**  
Admitted to the ENS through a competitive entrance examination.  
Four-year diploma completed in three years (accelerated track).
- 2026–2027 Research Year in Machine Learning**, *École Normale Supérieure Paris-Saclay*, France  
Final year of the ENS Diploma (research track).
- 2025–2027 MSc in Mathematics**, *École Normale Supérieure de Rennes*, France  
M2 Fundamental Mathematics. Highest honors (17/20 in the first semester).  
M1 Fundamental Mathematics. Highest honors (17.6/20). Rank: 5/100+.
- 2024–2025 BSc in Mathematics, Computer Science minor**, *École Normale Supérieure de Rennes*, France  
L3 Fundamental Mathematics. High honors (15.4/20).
- 2022–2024 BSc in Mathematics, Computer Science minor**, *University of Rennes*, France  
L2 Mathematics. Highest honors (18/20). Rank: 3/100+.  
L1 Mathematics. Highest honors (18.5/20). Rank: 1/100+.

## Publications and Preprints [\[More details\]](#)

- [1] **A. Autran**, Singular Tracking of Fractional Brownian Motion under Proportional Costs, preprint, 2026. Submitted.
- [2] **A. Autran**, Improved Wasserstein Estimates for Jump-Diffusion Perturbations under Semigroup Smoothing, preprint, 2026. Submitted.
- [3] **A. Autran**, Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem, preprint, 2026.
- [4] **A. Autran et al.**, Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles, *Applied Sciences*, 2026.

## Experiences [\[More details\]](#)

- Sep 2026–Jan 2027 **Machine Learning Researcher**  
High-Dimensional Density Analysis via Local Unimodality.  
Supervised by Argyris Kalogeratos (Senior Researcher) at Centre Borelli, ENS Paris-Saclay, France.
- May–Aug 2026 **M1 Research Internship**  
Optical Modeling of VO<sub>2</sub>-Based Thermochromic Oxide Nanocomposites.  
Supervised by Nguyen Ngoc Duy (Prof.) and Amaury Baret (PhD Student) at the Univ. of Liège, Belgium.
- Apr–May 2026 **Research Internship**  
Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems.  
Supervised by Martin Gander (Prof.) and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.
- Nov–Dec 2025 **M2 Research Seminar**  
Point Processes and Stein’s Method.  
Supervised by Jean-Christophe Breton (Prof.) at the University of Rennes, France.
- Jun–Aug 2025 **L3 Research Internship**  
Lattice-Based Post-Quantum Cryptography.  
Supervised by Pierre-Alain Fouque (Prof.) and Aurore Guillevis (Research Scientist) at Inria, France.
- May–Jun 2025 **Research Internship**  
Algebraic Counting of the Real Roots of Polynomials.  
Supervised by Marie-Françoise Roy (Prof. Emeritus) at the University of Rennes, France.

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## Other Activities [\[More details\]](#)

- 2024–2025 **ENS Oral Examinations**  
Supervision of second-year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures (Ulm, Paris-Saclay, Rennes, and Lyon).
- 2022–2025 **Teaching in Mathematics**  
Tutoring for final-year high school students at Lycée Chateaubriand in Rennes, as well as first- and second-year BSc students at the University of Rennes.
- 2023–2024 **BSc Student Representative**  
Representative on the council of the Mathematics Department of the University of Rennes.
- Before 2022 **High School Competitions**  
Participation in the Concours Général in Mathematics, Physics, and Philosophy, as well as the National High School Mathematics Olympiad, at Lycée Thiers in Marseille.

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## Computer Skills

**Programming:** Python (NumPy, pandas, scikit-learn, PyTorch), C/C++, Java, MATLAB, R.

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## Languages

**French:** native language.      **English:** professional working proficiency.

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## Sports and Interests

**Sports:** running, cycling, strength training.  
**Interests:** fantasy and science fiction literature; chess.

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## Academic References

**François Bolley** Email: [francois.bolley@ens-rennes.fr](mailto:francois.bolley@ens-rennes.fr).  
Full Professor and Director of the Mathematics Department at ENS de Rennes.

**Martin Gander** Courriel : [martin.gander@unige.ch](mailto:martin.gander@unige.ch).  
Full Professor, Section of Mathematics, University of Geneva.